

XGboost-Ampy: Identification of AMPylation Protein Function Prediction Using Machine Learning

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Abstract A developing post-translational modification known as AMPylation involves the formation of a phosphodiester bond on the hydroxyl group of threonine, serine, or tyrosine. Adenosine monophosphate is covalently attached to the side chain of an amino acid in a peptide during this process, which is catalyzed by AMPylation. We used AMPylation peptide sequence data from bacteria, eukaryotes, and archaea to train the models. Then, we compared the results of several feature extraction methods and their combinations in addition to classification algorithms to obtain more accurate prediction models. To prevent additional loss of sequence information, the PseAAC feature is employed to construct a fixed-size descriptor value in vector space. The basic feature set is received from 2nd features extraction method. All of this was accomplished by deriving the protein characteristics from the evolutionary data and sequence of the BLOSUM62 amino acid residue. The eXtreme Gradient Boosting (XGBoost) technique was used to create a novel model for the current study, which was then compared to the most popular machine learning models. In this research, we proposed framework for AMPylation identification that makes use of the XGBoost algorithm (AMPylation) and sequence-derived functions. XGBoost -Ampy has an accuracy of 86.7%, a sensitivity of 76.1%, a specificity of 97.5%, and a Matthews's correlation coefficient (MCC) of 0.753 for predicting AMylation sites. XGBoost -Amp, the first machine learning model developed, has shown promise and may be able to help with this problem.

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1 Introduction

Translational versus post-translational Earl Stadtman et al., who coined the term "AMPylation" from "adenylylation," found that a tyrosine residue in *Escherichia coli* glutamine synthetase was changed with AMP in the 1960s [1]. Proteins can be altered by enzymes or chemicals after they have been generated by the ribosome, a process known as post-translational modification (PTM). Protein posttranslational modifications (PTMs) occur when a translated protein undergoes cleavage, covalent alteration, or degradation [2] [3]. Protein dynamics and elasticity, for example, are only two examples of how these alterations shed light on a wide range of cellular functions and biological processes. For example, a protein undergoes proteolysis, oxidation, or degradation because of a change in its amino acid sequence, covalent modification, or degradation [4] [5]. These alterations provide insight into a wide range of protein-based cellular functions and biochemical pathways [6].

Protein structure, electrophilicity, and interaction are all influenced by them. To initiate a functional modification in their state, such as signaling or catalytic activity, PTMs can steer protein folding toward certain subcellular compartments, interacting ligands, or other proteins [7]. There have been a lot of PTMs found so far [8]. AMPylation refers to the covalent addition of AMP to proteins or peptides [9] - [10]. Only FIC domain proteins, which seem to be conserved and found in organisms ranging from bacteria to humans, have been used in the research. These enzymes mediate signaling in both bacteria and eukaryotes by binding AMP to Rho-family GTPases [11]. During the AMPylation process, adenosine monophosphate (AMP) is chemically linked to a protein's amino acid side chain. During walking, a nucleoside monophosphate nucleotide (i.e., adenylic acid) forms a phosphodiester link with a hydroxyl group of an AMPylated molecule [12]. Enzymes known as AMPylators are required to catalyze this reaction. The amino acids threonine (T) and tyrosine (Y) are particularly prone to AMPylation, while serine (S) is also a common target. This post-translational modification (PTM) is more pervasive than previously appreciated, as evidenced by recent proteomics studies, and is quickly becoming an important regulatory mechanism in both eukaryotic and prokaryotic cells. It controls many biological processes, from sickness in different animal species to regulation of bacterial nitrogen metabolism to signaling pathway regulation [13] - [14]. The regulation of neurodevelopment and neurodegeneration by AMPylation has also been established [15]. Methods used to identify PTM sites need a lot of time, money, and effort to implement. Because of this, many computer analyses [16] [17] have been written to predict where PTMs will be.

The most frequent and reliable type of AMPylation involves the formation of a phosphodiester link on the hydroxyl group of threonine, serine, or tyrosine. Adenosine Monophosphate (AMP) is covalently joined to the side chain of an amino acid in a protein molecule during the AMPylation process. Adenosine monophosphate nucleotide's phosphate group and the hydroxyl group of the molecule being AMPylated form a phosphodiester bond during the process of AMPylation (i.e., adenylic acid). The enzymes that can catalyze this reaction are known as AMPylators. AMPylation typically targets the amino acids threonine (T) and tyrosine (Y), while this PTM can also occasionally be seen in serine (S). As a result, numerous research has suggested employing quick and economical computational algorithms to anticipate PTM sites. Khater and Mohanty developed a computational procedure for identifying AMPylation domains and classifying them into different functional subfamilies such as those that catalyze AMPylation, deAMPylation, phosphorylation, and phosphocholine transfer in 2015. They did this using SVM and HMM [18] [19] [20]. Ghulam, Ali, et al. has been field-tested by extreme gradient boosting superfamily of cytoskeletal motor protein sites prediction method (XGBoost) for dynein protein's function predictions [21]. In 1799, Smith and Jones collaborated to investigate ubiquitination sites is often time- and labor-intensive, it is essential to create trustworthy predictions. In this study, we suggested a multi-view feature-based computational method for predicting ubiquitination sites called Ubipro-XGBoost [21]. Ample evidence [23] exists based on the BLOSUM vector score matrix, we extract IgG characteristics using the reduced feature dimension features that were chosen. Extreme Gradient Boosting is an ensemble learning technique we've created (XGBoost). It is [24] now generally accepted disease variation and new molecular correlations between genetic mutations, we carried out a pathway-based investigation. On the basis of shared gene interactions among illness-pathways, we created a biological network, and then we used network analysis to try and understand how a disease develops. In recent years, researchers have become increasingly interested in pathogenicity efficient impact. To create a "one-stop" conceptual model for the accurate determination of pathogenic potential and uses the best of options, [25] created VPatho, a two-stage deep-learning-based computational solution. Numerous computer algorithms for CPP prediction were developed until date. However, the present machine learning (ML) methods cannot differentiate between CPPs and their uptake efficiency. This research sought to construct a two-layer deep learning framework called DeepCPPred for identifying CPPs and peptide uptake efficiency [26]. The prediction of ACPs by empirical methods is costly, time-consuming, and frequently yields inaccurate results. It is in high demand to identify ACPs using sophisticated algorithms. In this article, ACP-2DCNN, a unique deep learning-based technique for enhancing the prediction of anticancer peptides [27]. Numerous computational studies have been conducted to distinguish illness nsSNPs from neutral ones, but there is still room for improvement in terms of overall prediction performance. Multi-scale convolutional neural network is a novel deep learning model proposed in this study (MSCNN) [28]. In this study, [29] suggest a deep learning approach based on a two-dimensional neural network (2D-CNN-PSPD). Consequently, the development of an exact technique for predicting its roles is a crucial step in the study of viral diseases and pathways. Numerous human disease pathways, including cancer pathways and signaling pathway-related diseases, have resulted

in the inactivation of PSPD. This investigation focuses on the molecular mechanisms that allow these various and diverse functions. Consequently, it is essential to increase the accuracy of immunoglobulin classification by employing efficient disease research techniques. Based on the BLOSUM vector score matrix, we use immunoglobulin features with reduced feature dimension to extract immunoglobulin features. We've developed a method called Extreme Gradient Boosting for ensemble learning (XGBoost) [30].

This paper presents a new framework for predicting protein subcellular localization by merging first pseudo-amino acid composition (PseAAC) features extraction method and second evolutionary data and sequence of the BLOSUM62 amino acid residue extraction method. BLOSUM62 has had a lot of success as a superb classifier and feature extractor, notably in the field of image recognition. Meanwhile, while adding new models, the gradient descent approach is used to reduce the loss. XGBoost -Ampy has an accuracy of 86.7%, a sensitivity of 76.1%, a specificity of 97.5%, and a Matthews's correlation coefficient (MCC) of 0.753 for predicting AMPylation sites. XGBoost -Ampy, the first machine learning model developed, has shown promise and may be able to help with this problem. In this investigation, 10-fold validation distances between the tips of atoms in different residues are used to describe residues that are close together. The outcomes demonstrated that the XGBoost model outperformed the other models with accuracy of 86.7%, and ROC (AUC) 0.886% for predicting AMPylation sites. Our findings show that it is possible to screen for antihypertensive peptides generated from eating a diet high efficiency and high performance by employing the XGBoost algorithm as a novel auxiliary tool.

2 Material and Methods

It is possible to build a novel machine learning model for protein association prediction by using the XGBoost AMPylation peptide protein sequence. A two-step function selection technique is implemented in order to remove the unnecessary functionalities from the model before the model is constructed. XGBoost AMPylation is then compared with two functionality encoding models, and the results are used as inputs for two machine learning classifications. We can also develop hybrid paces by combining various feature space combinations. For this purpose, 10-fold cross validation tests are also carried out. As shown in Figure 1. In this study, we employed the Kielkowski et al [31].

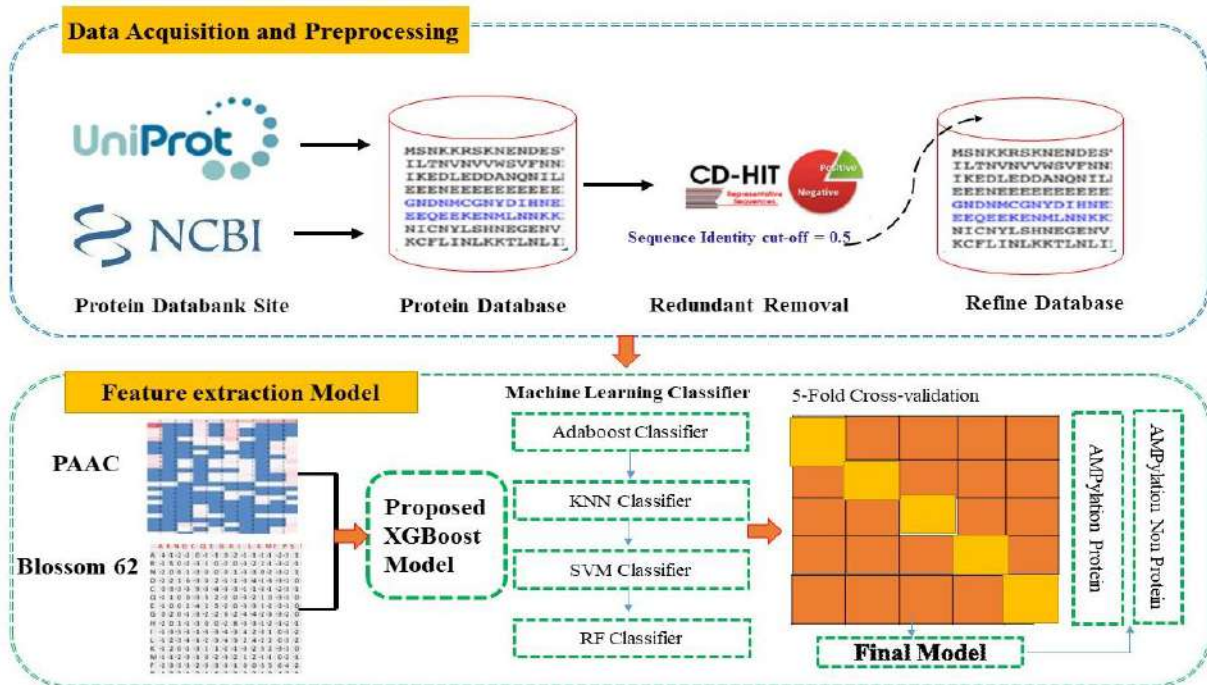


Figure 1: The XGBoost AMPylation protein framework model.

Benchmark dataset, which included 250 non-AMPylation peptide protein sequence and 153 AMPylation peptide protein sequence. The AMPylation peptides in this dataset were derived from XGBoost, while the 206 non-AMPylation peptides identified by Wang et al., [32] were derived from UniProt database to prevent the classifier from deviating, CD-HIT was used to exclude peptides with greater than 50% similarity from the data set. The same benchmark data set was used by Sieber, S et al. As well shown Table. 1.

Table 1. Collected benchmark datasets cross-validation

Protein	Original data	Similarity <40%	Cross-validation
AMPylation protein	165	153	153
Non - AMPylation protein	276	250	250
Total	441	403	403

2.1 Feature Extraction Strategies

Machine learning systems are unable to recognize the peptide sequences right away. As a result, it is necessary to convert the strings representing peptide sequences into numerical characteristics [33]. To create computational predictors, feature extraction approaches are critical [34]. We used two different feature extraction algorithms in this paper, including amino acid composition (PseAAC) and the BLOSUM62 matrix. Several feature extraction methods may be included in each strategy. These ideas have been implemented through the use of functionalities [35].

2.2 Feature encoded by Pseudo-Amino Acid Composition

To create a discrete number of each protein sample is PseAAC's (Pseudo Amino Acid Composition) fundamental tenet. This model accounts for the locations of amino acids in a peptide sequence as well as their frequencies. To compensate for the loss of sequence information, the PseAAC feature creates a fixed size descriptor value in vector space, which is then utilized to advance systematic evolution. An approach comparable to PAAC was also utilized in this paper. Chou et al. Proposed composition model for pseudo-amino acids amphiphiles (APAAC). The hydrophilic and hydrophobic characteristics of amino acids are taken into account in the model. The correlated factors in the sequence order are described below.

$$\theta_d = \frac{1}{N-d} \sum_{i=1}^{L-d} \Theta(R_i, R_{i+d}), d=1,2,\dots,nlag \quad (1)$$

Pseudo-Amino Acid Composition can be formulated as $P = [p_1, p_2 \dots p_{20}, \dots, P_2 P_{20+nlag}]$

$$p_u = \begin{cases} \frac{f_u}{\sum_{r=1}^{20} f_r + w \sum_{d=1}^{nlag} \theta_d}, u=1,2,\dots,20 \\ X_r = \frac{w \theta_u}{\sum_{r=1}^{20} f_r + w \sum_{d=1}^{nlag} \theta_d}, u=21,22,\dots,20+nlag \end{cases} \quad (2)$$

2.3 Coevolution correlations can be extracted from numerous sequence alignments.

The coevolution correlations are extracted using 403 AMPylation proteins sequence -A BLOSUM62 in this work. Every family has at least a thousand sequences and one structure that has been determined experimentally. Each BLOSUM 62 computation takes into account only "ungapped" places. In this case, an "ungapped" position is one with fewer than 10% gaps. In this work, we leverage mutual information to infer relationships between species' evolutionary histories. An unrepresentative sample of dependent substitutions could be generated by using a method like XGBoost, which attempts to predict residue interactions. When doing sequence matching, it is important to take into account the transitive correlations that come from shared knowledge, such as knowledge about how different substitutions depend on each other. In a BLOSUM62, reciprocal information between two locations is defined as:

$$L_{ij} = \sum_{A_i, B_j}^a f(A_i, B_j) \log \frac{f(A_i, B_j)}{f(A_i) f(B_j)} \quad (3)$$

where q is the total number of amino acid types and n is the number of deficiencies. Since there are a lot of sequences in the

XGBoost dataset, we can utilize frequency to assess the likelihood of events. The frequency of a single amino acid of type A at position I can be computed using the formula $f(A, I)$, which is the frequency with which amino acids A and B are found together at positions I and j . $F(A, i, B, j)$ is the frequency with which two different types of amino acids are observed together at positions I and j , the highly connected sets of items.

2.4 Selecting correlated positions with observed proximate residues.

Using the complete structural information in the training data set, we can eliminate a large number of false positives by focusing on the closeness of observed residues [36] [37]. In this investigation, 10-fold validation distances between the tips of atoms in different residues are used to describe residues that are close together. The use of advanced atoms facilitates the identification of proximal residue groups with large sequence separations that are most susceptible to interaction.

2.5 Derive a new matrix from contextual substitutions.

Surrogate scores are components of an amino acid surrogate matrix. Each protein family shows evidence of evolutionary association, as measured by the evolutionary computation. The correlation of coevolution is observed in each protein family, depending on the results of the coevolution computation. The majority of amino acid mutations are reliant on the structural context in which they occur. We were able to include information about how substitutions depend on each other in a new set of substitution scores by taking the frequency of substitutions from the previously chosen places that were related.

$$s(A, B) = \lambda \log \frac{f(A, B)}{f(A) f(B)} \quad (4)$$

where $f(A, B)$ and $f(A, B)$ represent the probabilities of occurrence of each individual amino acid type and the combined probabilities, respectively. Based on the training dataset, is a scalar factor that has been tuned to deliver the greatest homolog detection performance (402 AMPylation domains). To optimize, a semi-heuristic technique is applied. Iteratively, the study begins with a coarse set of values covering a large range, then it is refined over a narrower range. In the first cycle, we go through the values from 1.0 to 5.0 in 0.5 increments. When the stepping interval is smaller than 0.1, the operation ends. As a result, in the final ProtSub matrix, 2.4 is assigned.

2.6 Proposed XGBoost Model

With the BLOSUM62 and -XGBoost show for protein subcellular area forecast, Figure 1 delineates the by and large structure of the pseudo-amino corrosive composition (PAAC). The model's input may be a one-dimensional vector built utilizing location-specific scoring frameworks (Blousum62) and proteins interaction scoring networks collected from semantic likenesses between STRING and GO watchwords [38] [39]. A protein can be communicated as an $L \times 1$ vector (L being the number of groupings within the preparing set), whereas simple picture information may be a one-dimensional "picture" with 1 channel. As a result, the input could be a $L \times 1$ matrix.

After getting the correct include illustrations by the prepared XGBoost, related with the classic XGBoost. We conducted all analyses using BLOSUM62 and PAAC-XGBoost demonstrate replaces the vector matrix of BLOSUM62 62 and PseAAC with XGBoost to foresee the localization of subcellular of proteins, which empowers highlights consequently gotten from input and gives more exact and proficient classification.

2.7 Classification

The process of identifying proteins linked to diseases can be viewed as a binary classification problem, in which proteins are either "in" or "out" depending on the context. We used well-known classifiers like AdaBoost, SVM, Random Forest (RF), and KNN to separate proteins that are linked to infectious diseases from other proteins.

2.8 K-Nearest Neighbor (KNN)

A fundamental supervised learning method that may be applied to both classification and regression is the K-Nearest Neighbor (KNN) Method. Because it uses every piece of data for training and prediction, it has earned the moniker "lazy learner" [40] [41]. At the outset, we need to decide on a value for k , where n is the total number of records. A distance metric function of choice is then used to determine how far apart each row of test and training data is (use Hamming distance for categorical data). With the distance values computed, it now arranges them from closest to farthest. Once the array has been sorted, the top k rows are selected, and the test point is assigned a class based on the class with the highest percentage of votes (the majority of rows) [42].

2.9 Random Forest (RF) Classifier

As an ensemble method, the Random Forest (RF) technique takes the combined results of several different tree-based classifiers [43]. By averaging the outcomes of models trained on separate sub-samples of the dataset, random forest (RF) can improve model performance while preventing over-fitting. Due to the intricacy of its models, RF is both computationally expensive and time-consuming compared to other methods of making forecasts. Another well-liked strategy for developing clinical prediction models is RF [44] [45].

2.10 Performance measures and results

The main goal of this work was to determine whether a sequence was AMPylation protein or not, hence we used the terms "Positive" to denote AMPylation protein and "Negative" to denote non-AMPylation protein. Five-fold cross-validation was used to train models for each dataset. We used 5-fold cross-validation to determine the best model to use with each dataset and then used hyper-parameter optimization to zero in on that model. Finally, the present model's prediction capacity was tested using an external dataset.

Chou's symbols, originally developed for studying protein signal peptides, served as the inspiration for a set of four simple metrics [45]. Chou's criterion, which has been used in many bioinformatics studies [46] [47], was another metric we used to evaluate the method. Chou's criterion, which has been used in many bioinformatics studies [48] [49], was another metric we used to evaluate the methods. Only single-label systems (where each sample only belongs to one class) can use the intuitive metrics generated from Chou's symbols or the standard metrics extracted directly from mathematics textbooks [50]. Accordingly, the domains of system biology, system medicine, and biomedicine increasingly employ multi-label systems (where a sample may belong to multiple classes at once), which require an entirely different set of metrics. Sensitivity, specificity, accuracy, and Matthews's correlation coefficient (MCC) were calculated using the following formula: The four outcomes are true positive (TP), false positive (FP), true negative (TN), and false negative (FN).

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (5)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (6)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \quad (7)$$

$$\text{MCC} = \frac{TP * TN - FP * FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (8)$$

3 Result and Discussion

The study's quality and dependability of research modelling approaches is a critical aspect. In the results and discussions section, we first built an experiment by examining data performing computations and making performance comparisons based on different metrics score as shown in table 2.

Table 2. XGBoost Identifying optimum parameter for various models.

Classifier	ACC	Precision	Sensitivity	Specificity	MCC	ROC-AUC
B62 _AMPylation	0.867%	0.876%	0.76%	0.975%	0.753%	0.886%
PseAAC _AMPylation	0.862%	0.772%	0.755%	0.970%	0.733%	0.874%

The analysis was based on BLOUSM62 features using with XGBoost algorithm has an accuracy of 86.7%, a sensitivity of 76.1%, a specificity of 97.5%, and a Matthews's correlation coefficient (MCC) of 0.753 and ROC (AUC) 0.886% for predicting AMPylation sites. The analysis was based on PseAAC features using with XGBoost algorithm has an accuracy of 86.2%, a sensitivity of 0.755%, a specificity of 0.970% and a Matthews's correlation coefficient (MCC) of 0.733% and ROC (AUC) 0.874% for predicting AMPylation sites. It was possible to obtain 5-fold cross-validation of various prediction models with various properties. We used 5-fold cross-validation, model performance comparisons on the difference between positive and negative data sets are the outcomes. Our forecast performance was relatively good according to the statistical study with an overall accuracy of 0.867%.

This suggests that the ROC(AUC) is approximately achieved based on Bloosm62 features 0.91% which indicating that our model's estimation is quite good. This suggests that the ROC(AUC) is approximately achieved based on PseAAC features 0.87% which indicating that our model's estimation is quite good. The AMPylation proteins exhibit strong ROC curve performance

(AUC=0.91 percent) using a Blossom62-extracting feature model, and the ROC curve also performs well. Using another feature model, PAAC the AMPylation proteins, we were able to acquire a high ROC cure performance for our model as shown in Figure 2.

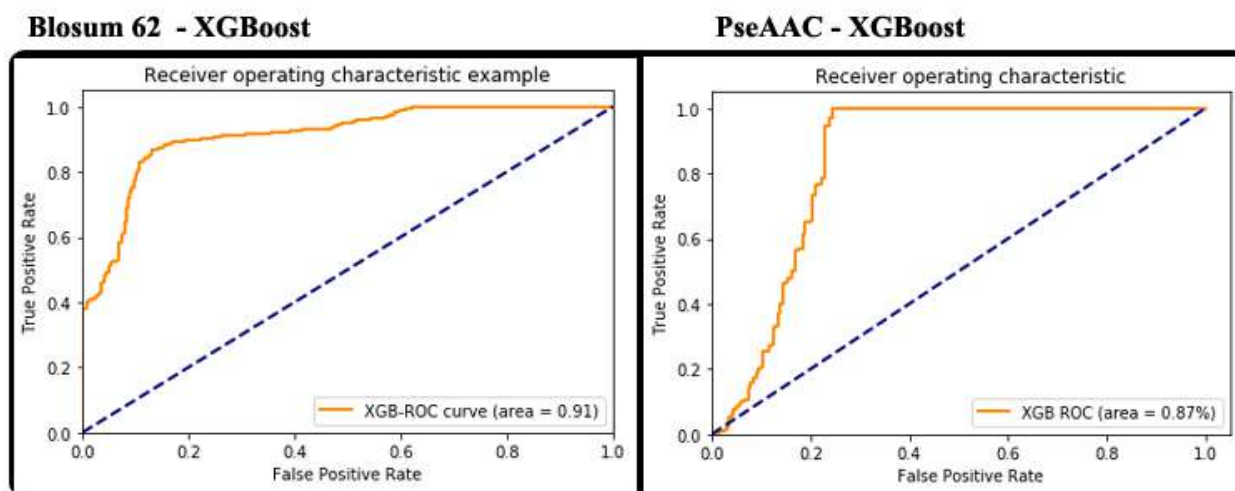


Figure 2. Proposed model ROC-AUC score predictive models.

3.1 Comparison with different machine learning techniques

We looked assessed how well different machine learning classifiers performed at finding AMPylation proteins. To test the model, we employed four different classifiers: AdaBoost [51] (i.e., closest neighbor (KNN), Gaussian, Random Forest, and support vector machine (SVM), and compared the results of XGBoost with theirs. In all of the trials, we used the optimal settings for all of the classifiers to ensure a fair comparison. Table 3 compares the performance of our method to that of different machine learning algorithms. Using the same experimental setting, it can be seen that our XGBoost performed better than the other classic machine learning algorithms. When using the independent dataset, our XGBoost algorithm outperformed other algorithms.

Table 3. Comparison of BLOSUM 62 and PAAC on AMPylation dataset with other methods.

	Classifier	Acc	Prec	Sen	Spec	MCC	ROC-AUC
BLOUSM62	AdaBoost	0.803%	0.781%	0.77%	0.831%	0.6198%	0.863%
	KNN	0.566%	0.560%	0.60%	0.526%	0.1347%	0.599%
	SVM	0.607%	0.646%	0.49%	0.718%	0.2242%	0.637%
	RF	0.800%	0.785%	0.77%	0.825%	0.6164%	0.864%
	XGBoost	0.867%	0.876%	0.76%	0.975%	0.753%	0.886%
PAAC	AdaBoost	0.817%	0.738%	0.77%	0.865%	0.6337%	0.856%
	KNN	0.825%	0.836%	0.79%	0.860%	0.6683%	0.886%
	SVM	0.857%	0.763%	0.76%	0.955%	0.6976%	0.887%
	RF	0.865%	0.777%	0.76%	0.975%	0.7414%	0.872%
	XGBoost	0.862%	0.772%	0.75%	0.970%	0.733%	0.874%

Overall, these findings demonstrate the intuitive validity of Figure 3. AMPylation proteins are involved in the process. The precision of AUC values is reached using ROC curves similar to previous ways; 93.53%, 88.13 %, 83.21 % F1 85.31% and ROC AUCs. The procedure's output is the ROC-AUC. 92.62 % specificity 93.53 %, as seen in Figure 3. In this experiment, we evaluate our suggested AdaBoost model against three other popular classification algorithms: the nearest neighbor classifier (KNN), the random forest classifier (RFC), and the support vector machine (SVM).

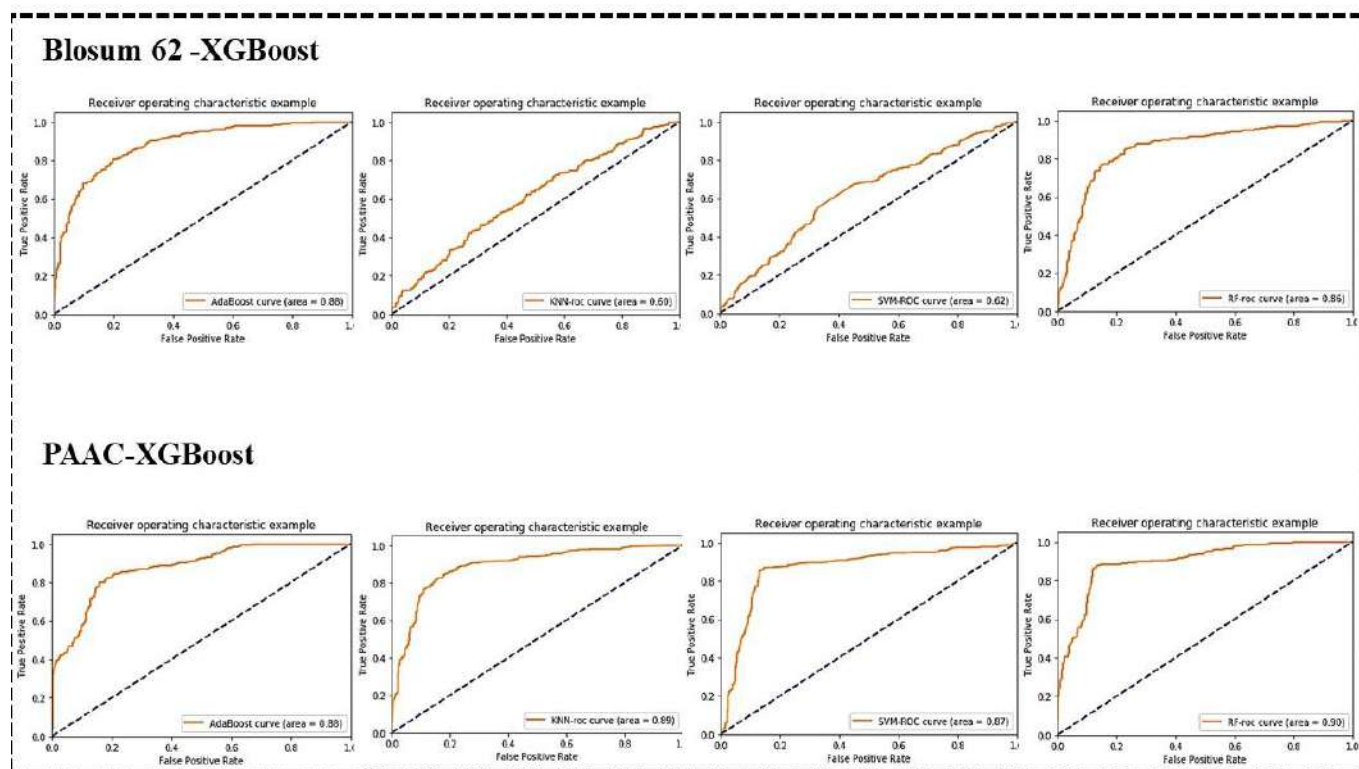


Figure 3: A comparison of the proposed model to some other classifiers

3.2 Identifying a Comparison of combined four Feature Extraction Approaches

BLOSUM62 and PAAC (AUC=0.91) are two ROC(AUC) curves that suit AMPylation proteins and contain ROCs according to one feature extraction approach. According to these ROC(AUC) outputs of AdaBoost and SVM and KNN and Random Forest (XGBoost) as shown in figure 4, are better than other approaches.

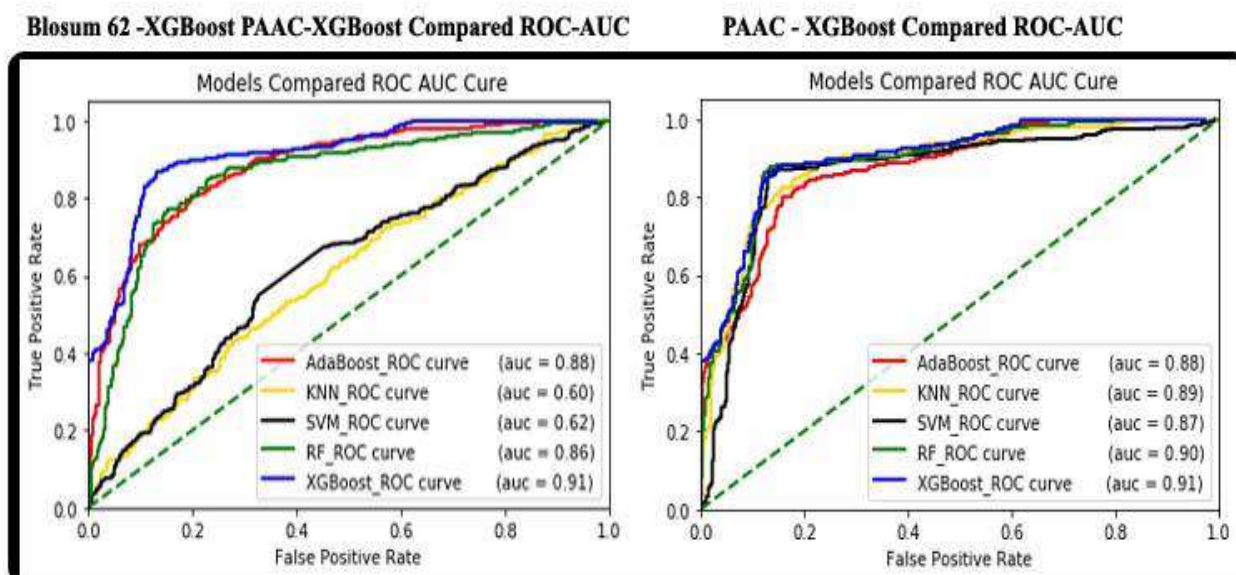


Figure 4: ROC-AUC score result for the proposed two-model

3.3 Performance comparison of different Machine learning classifiers

Blossum62 and PAAC classifier accuracy is best with XGBoost, followed by four classification algorithms, including the first XGBoost. The Blossum62 and PAAC models then received a precision score, and the Blossum62 model received a third accuracy score. Results from our experiments show that XGBoost classification is the most accurate overall, followed by AdaBoost, and then KNN, RF, and SVM classification. The Blossum62, PAAC model's third-best classifier is linear discriminant analysis, which is just one of the many powerful statistical methods that can be used. PAAC model, was obtained, as shown in Figure 5.

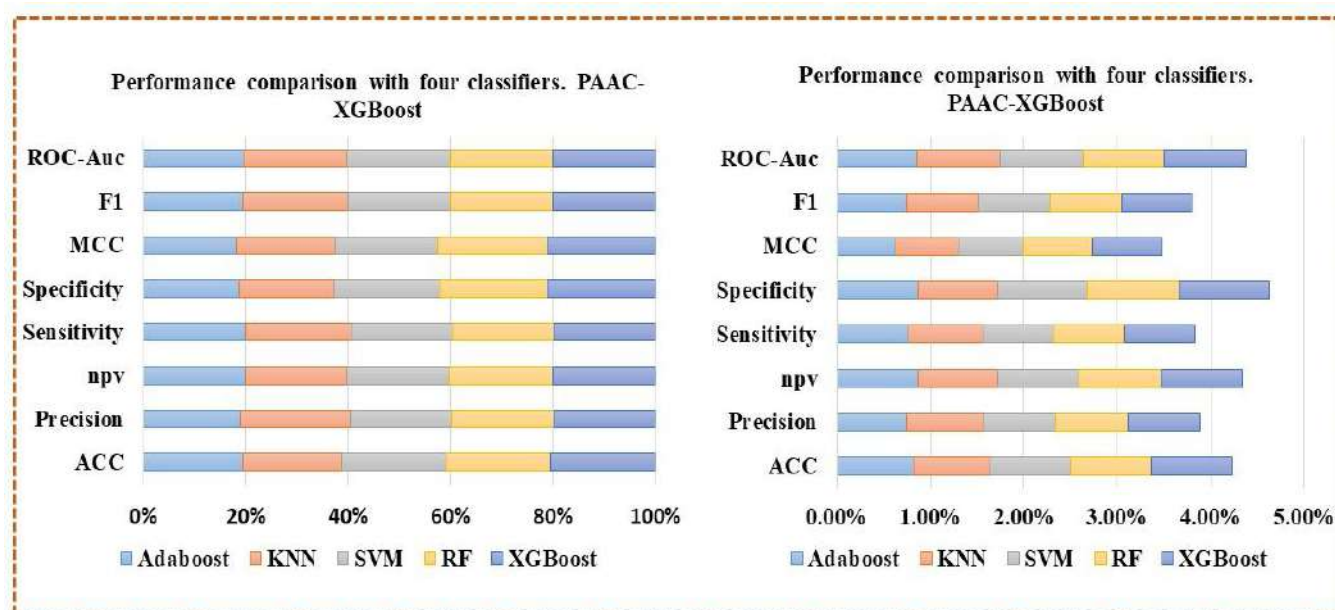


Figure 5. Performance comparison with four classifiers

Conclusion:

The function of AMPylation is still developing, and there are still many unanswered concerns regarding the diversity and nature of this post-translational modification. The function of AMPylation in endogenous eukaryotic signaling comes first. Several eukaryotic Fic proteins' domain architectures, as previously demonstrated, points to membrane localization and a function in metabolism or stress response. The performance of four distinct machine learning classifiers—AdaBoost, KNN, SVM, and RF—using an object-based classification method to categories extremely high-resolution photos was evaluated and contrasted in this study.

Each classifier (with the exception of XGBoost) had its tuning parameters examined to see how they impacted classification accuracy with varying training sample sizes. In this study, AdaBoost and KNN were found to be superior to SVM and RF when it came to classifying urban terrain. Each of the two classifiers has its advantages and disadvantages; therefore, choosing the optimal one may depend on the specifics of a given situation. While XGBoost has the potential to achieve high accuracy with a small number of training samples, finding the optimal values for its tuning parameters may be an inherently subjective and time-consuming process. While SVM does not require any tuning parameters, it does require a large number of training instances to achieve fairly acceptable accuracy.

Tuning parameter settings had the greatest impact on SVM, followed by AdaBoost, while KNN was somewhat immune to their influence. While there was some variation in the optimal values for the tuning parameters based on the sizes of the training sets, commonalities did emerge. The best average accuracy was typically achieved with SVM parameters C between 1,000,000 and 100,000,000 and gamma between 0.00001 and 0.001. The highest depth of the classification tree discovered to provide the best XGBoost classification accuracy was between five and eight. KNN worked best with a K value of two or three. For predicting AMPylation sites, XGBoost-Amp also considerably outperforms popular machine learning models like Support Vector Machine, K-nearest Neighbor, and Random Forest. The little sample size that is available makes it difficult to make predictions with high accuracy. We intend to expand the database for AMPylation and add more AMPylation sites to the dataset as we continue to optimize our work. Additionally, we seek to enhance the performance of our predictor by utilizing additional XGBoost-Amp architectures and feature sets.

Credit Author Statement

Rahu Sikander made an equal contribution to this article. The idea for the analysis of the manuscript came from **Ali Ghulam**. The original draft was written by **Zar Nawab Khan Swati**. **Rahu Sikander**, and **Jawad Usman Arshad** was instrumental in developing the strategy and the software. After receiving feedback, **Muhammad Shahid Malik**, **Nauman khan** and **Muhammad Sohail** edited the manuscript to improve the English language and flow. The final manuscript has been read and approved by all writers.

Compliance with Ethical Standards:

It is declare that all authors don't have any conflict of interest. It is also declare that this article does not contain any studies with human participants or animals performed by any of the authors. Furthermore, informed consent was obtained from all individual participants included in the study.

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